

Scientists designed an experiment to investigate DNA replication.

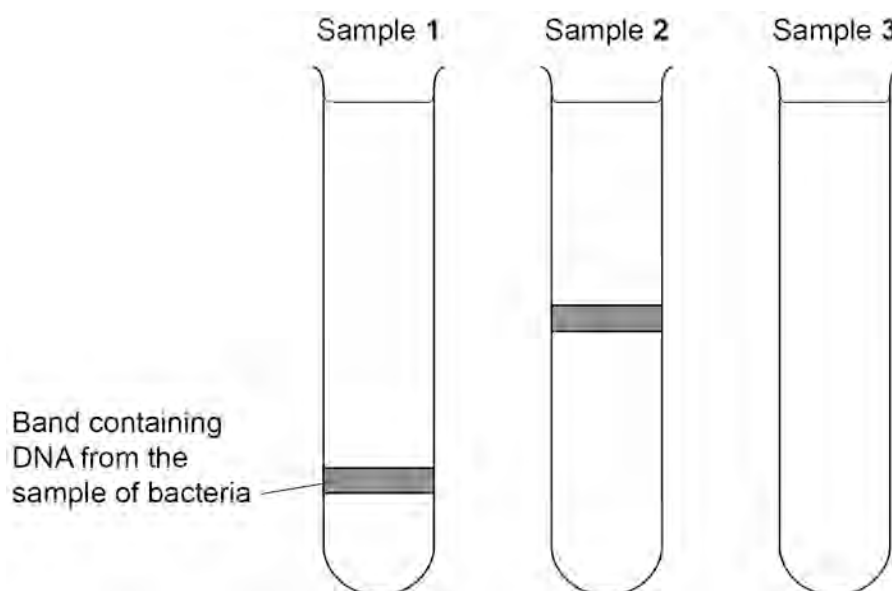
They grew bacteria for several generations in a nutrient solution containing a heavy form of nitrogen (^{15}N). They obtained DNA from a sample of these bacteria (sample 1).

The scientists then transferred the bacteria to a nutrient solution containing a light form of nitrogen (^{14}N). They allowed the bacteria to grow and divide twice. After each division, they obtained DNA from a sample of bacteria (sample 2 and sample 3).

They suspended DNA from each sample of bacteria in a solution in separate tubes. They spun these tubes in a centrifuge at the same speed and for the same time. Based on its density, DNA settled as bands in each tube.

Figure 5 shows the scientists' results.

Figure 5



0 3 . 2 Explain the position of the DNA band in sample 2 in **Figure 5**.

[2 marks]

0 3 . 3 Complete sample 3 in **Figure 5** to show the result you would expect after two divisions.

[1 mark]

8

Turn over ►

